

Sadeep Ariyaratna

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EDUCATION

South Westphalia University of Applied Sciences

MSc. in Systems Engineering and Engineering Management

Soest, Germany

Apr. 2023 – Present

General Sir John Kotelawala Defense University

BSc. (Hons.) in Mechatronics Engineering

Ratmalana, Sri Lanka

2022

EXPERIENCE

Intern – E/E Architecture

CLAAS E-Systems

Oct. 2024 – Present

Dissen, Germany

- Modelling system architecture in Enterprise Architect, focusing on system engineering concepts.
- Developing conceptual methods for improving model collaboration in MBSE.

Mechatronics Engineer

AtlasLabs (Pvt.) Ltd, Sri Lanka

Sept. 2022 – Feb. 2023

Colombo, Sri Lanka

- Engineered and tested mechanical components and systems from concept to launch with Fusion 360, reducing product development cycle time by 20%.
- Created manufacturable designs using Fusion 360 and validated them for Design for Manufacturing and Design for Assembly techniques.
- Reduced troubleshooting time by 35% by creating bill of materials and validation documents for inspections.

In-plant Trainee – Automation

Southwestern University

July 2021 – Feb. 2022

Georgetown, TX

- Created SolidWorks designs using Design for Manufacturing and Design for Assembly techniques.
- Designed and developed an attachment for sewing machines that reduced the time taken to cut the finished piece by 65%.
- Developed control systems for three ongoing projects to evaluate each conceptual method, using microcontrollers (Arduino, ATTiny84).

PROJECTS

Chess Game | Python, Reinforcement Learning

2024

- Developed a fully functional chess game with time management, move validation, move history and move notation features.
- Developed a reinforcement learning algorithm using DQN to train a chess agent.

IEEE BibTex Generator | Python, Package Development, Flask

2024

- Developed a Python package to simplify the creation of BibTex entries for sources used in LaTeX. The package can automatically generate the BibTex entry for a given DOI.

Remote Temperature Detection and Transmit | C, Arduino, MQTT, Embedded Programming

2024

- Developed an application to detect temperature at a remote location and transmit using a GSM module. The application was developed for STM32 platform using C. The application has the capability to store the temperature readings and transmits using the GSM module via MQTT.

Grid Solving with Reinforcement Learning | Python, Reinforcement Learning

2024

- Developed a custom grid environment and analyzing performance of Q-learning, Deep Q-learning and A2C algorithms using Pytorch, OpenAI Gym and StableBaselines3.

Prosthetic Elbow Concept | C++, Arduino, Embedded Programming, SolidWorks

2022

- Engineered a C++-based control system for precise control of an elbow joint using a brushless DC motor with in a prosthetic arm. Designed and optimized the intricate gear system for the joint utilizing SolidWorks, ensuring seamless integration and functionality.

TECHNICAL SKILLS

Languages: Python, C/C++, Arduino, MATLAB

Software: SolidWorks, Fusion 360, MATLAB Simulink, Autodesk Eagle, CubeIDE

Developer Tools: Git, Github, ADO

Libraries: Pytorch, Tensorflow, StableBaselines, OpenAI Gym